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Computer Vision Project 2

Text to Video Retrieval

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Abstract

Text-to-video retrieval aims to identify videos that are semantically relevant to a natural language query, a task challenged by the semantic gap between visual and textual modalities. Recent vision–language models such as CLIP enable effective cross-modal alignment through contrastive pretraining, making similarity-based retrieval feasible without task-specific supervision. This work presents a CLIP-based text-to-video retrieval system evaluated on the Microsoft Research Video to Text (MSR-VTT) dataset. Videos are represented by uniformly sampling frames and encoding them using a pretrained CLIP image encoder, followed by mean pooling to obtain a video-level embedding. Text queries are encoded using the CLIP text encoder, and retrieval is performed using cosine similarity in a shared embedding space. To improve retrieval accuracy at higher ranks, a two-stage retrieval strategy with cross-encoder reranking is employed. Experimental results indicate that the visual-only CLIP baseline achieves strong performance, while reranking provides modest but consistent improvements, highlighting the trade-off between retrieval efficiency and semantic precision.

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List of Acronyms

AP Average Precision

MeanR Mean Rank

MedR Median Rank

CV Computer Vision

mAP Mean Average Precision

R@K Recall at K

MSR-VTT Microsoft Research Video to Text

CLIP Contrastive Language-Image Pre-training

MEE Mixture-of-Embedding-Experts

CLIPBERT Clip Bidirectional Encoder Representations from Transformers

CLIP4Clip CLIP For Video Clip retrieval and captioning

CLIP2Video CLIP to Video-Text Retrieval

X-CLIP extentioned CLIP

CenterCLIP Token Clustering for Efficient Text-Video Retrieval using CLIP

CLIP2TV CLIP-based transformer-based method for video-text retrieval

CLIPPING Distilling CLIP-Based Models with a Student Base for Video-Language Retrieval

NarVid Narrating the Video

FAISS Facebook AI Similarity Search

L2-normalized Euclidean normalization

ViT Vision Transformer

GPU Graphics Processing Unit

Chapter 1

Background and Literature Review

Text-to-video retrieval aims to retrieve videos that are semantically relevant to a given natural language query. The task is challenging due to the temporal structure of video data, the semantic gap between visual and textual modalities, and the weak supervision typically available in large-scale video–text datasets. Early approaches relied on handcrafted multimodal pipelines and task-specific feature extractors, which limited scalability and generalization.

The emergence of large-scale vision–language pretraining models, most notably CLIP [1], marked a paradigm shift in multimodal retrieval. By learning aligned visual and textual representations through contrastive learning on massive image–text corpora, CLIP enabled efficient similarity-based retrieval without task-specific heads. Building upon this foundation, subsequent research has explored how CLIP can be transferred and adapted to the video domain, focusing on efficient video representation, temporal reasoning, cross-modal alignment, and system-level trade-offs between accuracy and computational cost. This section reviews representative methods along these dimensions in chronological order.

1.1 Literature Review

Pre-CLIP Approaches to Text-to-Video Retrieval

Before the advent of large-scale vision–language pretraining, text-to-video retrieval systems were constructed using heterogeneous expert models and handcrafted fusion strategies. Liu et al. introduced the Collaborative Experts framework [2], which represents videos by aggregating features from multiple pretrained expert models, including object recognition, action recognition, scene classification, audio analysis, speech recognition, and optical character recognition. A query-dependent gating mechanism dynamically weights each expert according to its relevance. While effective, this approach incurs high computational cost and architectural complexity due to its reliance on multiple task-specific models.

To address the challenge of incomplete and heterogeneous multimodal supervision, Miech et al. proposed the Mixture-of-Embedding-Experts (Mixture-of-Embedding-Experts (MEE)) framework [3]. MEE learns modality-specific embeddings and combines them

flexibly, enabling retrieval even when certain modalities are missing. Together, these pre-CLIP approaches highlight the importance of multimodal fusion while exposing the scalability limitations of expert-based pipelines.

Dual-Encoder and Sparse Temporal Modeling

A key conceptual transition occurred with the adoption of dual-encoder architectures, in which video and text are independently encoded into a shared embedding space and compared using similarity-based retrieval. This paradigm enables efficient large-scale retrieval while avoiding expensive cross-modal interaction during inference.

Lei et al. introduced Clip Bidirectional Encoder Representations from Transformers (CLIPBERT) [4], demonstrating that sparse temporal sampling can be sufficient for effective video–language learning. By processing only a small number of short clips per iteration, CLIPBERT significantly reduced computational cost while maintaining competitive performance. These findings motivated later CLIP-based approaches that prioritize efficiency over dense temporal modeling.

Large-Scale Vision–Language Pretraining

CLIP [1] fundamentally reshaped multimodal retrieval by introducing contrastive pretraining on over 400 million image–text pairs. Using dual encoders for images and text, CLIP learns aligned embeddings that enable zero-shot retrieval via cosine similarity in a shared semantic space. Although CLIP is trained on images rather than videos, its strong semantic alignment properties make it a powerful backbone for text-to-video retrieval, where video frames can be treated as image samples.

First Generation of CLIP-to-Video Transfer (2021)

CLIP For Video Clip retrieval and captioning (CLIP4Clip) [5] provided the first systematic study of transferring CLIP to text-to-video retrieval. The authors explored multiple frame aggregation strategies and similarity computation mechanisms, demonstrating that uniform frame sampling followed by mean pooling of CLIP embeddings achieves strong retrieval performance. This work established CLIP-based mean-pooled video representations as a simple yet competitive baseline, showing that sophisticated temporal modeling is not strictly necessary for effective retrieval.

Building upon this baseline, CLIP to Video-Text Retrieval (CLIP2Video) [6] explicitly addressed CLIP's lack of temporal awareness by introducing a Temporal Difference Block to capture motion cues and a Temporal Alignment Block to align video frames with textual phrases. By incorporating lightweight temporal reasoning, CLIP2Video demonstrated improved retrieval accuracy while retaining the benefits of CLIP-based representations.

Multi-grained Contrastive Learning and Fine-Grained Alignment (2022)

As CLIP-based baselines matured, subsequent work focused on improving cross-modal alignment through fine-grained interactions. extentioned CLIP (X-CLIP) [7] introduced multi-grained contrastive learning, jointly modeling video–sentence, frame–word, and cross-grained interactions. To effectively aggregate these similarities, X-CLIP proposed an Attention Over Similarity Matrix module that emphasizes informative frames and words while suppressing irrelevant ones. This work established multi-grained contrast as a key architectural direction for enhancing retrieval accuracy beyond coarse video–sentence alignment.

Efficiency-Oriented CLIP Architectures

Alongside accuracy-focused methods, several works emphasized efficiency and deployability. Token Clustering for Efficient Text-Video Retrieval using CLIP (CenterCLIP) [8] observed that CLIP's vision transformer produces redundant visual tokens across adjacent frames. By introducing segment-level token clustering, CenterCLIP reduced computation and memory usage while preserving semantic alignment.

CLIP-based transformer-based method for video-text retrieval (CLIP2TV) [9] decomposed retrieval into video–text alignment and video–text matching modules, combining embedding-level similarity with transformer-based matching and similarity distillation to handle noisy captions. Distilling CLIP-Based Models with a Student Base for Video-Language Retrieval (CLIPPING) [10] further explored parameter efficiency through knowledge distillation, transferring visual and cross-modal knowledge from a large CLIP-based teacher to a lightweight student model. These works collectively highlight a shift toward system-level optimization, balancing retrieval accuracy and computational efficiency.

Coarse-to-Fine Retrieval and Semantic Enrichment (2024–2025)

Recent research has increasingly adopted coarse-to-fine retrieval paradigms. Tian et al. proposed a two-stage retrieval framework in which coarse-grained video representations are used for fast candidate recall, followed by fine-grained reranking using text-gated interaction blocks [11]. This design reflects a growing consensus that heavy cross-modal fusion is unnecessary in early retrieval stages.

Another emerging direction focuses on enriching video representations using auxiliary textual signals. Hur et al. introduced Narrating the Video (NarVid) [12], which incorporates frame-level generated captions, referred to as narration, as an auxiliary modality. By leveraging query-aware caption filtering, dual similarity fusion, and hard-negative learning, NarVid demonstrated that language-augmented temporal reasoning can substantially improve retrieval robustness, while also highlighting the challenges posed by noisy generated text.

Chapter 2

Methodology and System Design

2.1 Dataset

The MSR-VTT (Microsoft Research–Video to Text) dataset is a large-scale benchmark designed to advance research at the intersection of video understanding and natural language processing. Introduced by Xu et al. in 2016, MSR-VTT was created to address the limitations of earlier video–language datasets, which were often small in scale, restricted to narrow domains, or lacked linguistic diversity. The dataset consists of 10,000 web video CLIP totaling 41.2 hours, collected from 257 popular video search queries spanning 20 diverse categories such as sports, cooking, news, music, and instructional content. Each video CLIP is annotated with approximately 20 natural language sentences, resulting in 200,000 video–sentence pairs and a rich vocabulary of over 29,000 unique words. This combination of scale, category diversity, and dense human annotations makes MSR-VTT particularly suitable for evaluating video–text tasks such as video captioning, text-to-video retrieval, and cross-modal representation learning, and it has since become one of the most widely used benchmarks in video–language research[13].

2.2 Multimodal Deep Learning Architecture

System Overview

The proposed system addresses the text-to-video retrieval task by learning a shared semantic embedding space between visual and textual modalities. Given a natural language query, the system retrieves semantically relevant videos by comparing independently encoded video and text representations using similarity-based retrieval.

The overall architecture follows a dual-encoder design. Video content is represented through frame-level visual embeddings extracted using a pretrained vision–language model, while text queries are encoded using a corresponding text transformer. Both encoders project their respective inputs into a common latent space, where similarity is computed via cosine similarity. This design enables efficient large-scale retrieval by allowing video embeddings to be precomputed and indexed in advance.

During training, the vision encoder is kept frozen, and only the text encoder is fine-

tuned using a contrastive learning objective to improve alignment between textual queries and video representations. At inference time, retrieval is performed using a single-stage embedding-based search over a Facebook AI Similarity Search (FAISS) index, ensuring scalability and low computational overhead. The system deliberately avoids explicit temporal modeling and cross-modal interaction during inference, prioritizing efficiency and robustness under limited computational resources.

Visual Encoder and Video Representation

Video representations are constructed using a frame-based encoding strategy. For each video, a fixed number of frames (8) are uniformly sampled across its temporal duration. Each sampled frame is independently processed by the image encoder of a pretrained CLIP model, producing a sequence of frame-level embeddings. Each sampled video frame is encoded using the CLIP image encoder, as shown in Equation 2.1.

$$\mathbf{v}_i = f_{\text{img}}(x_i), \quad i = 1, \ldots, N, \tag{2.1}$$

where x_i denotes the i-th sampled frame and $f_{\text{img}}(\cdot)$ represents the CLIP image encoder.

To obtain a single video-level embedding, the frame embeddings are aggregated using mean pooling seen in Equation 2.2.

$$\mathbf{v}_{\text{video}} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{v}_i, \tag{2.2}$$

where N denotes the number of sampled frames. The resulting embedding is Euclidean normalization (L2-normalized) before similarity computation. This aggregation strategy avoids explicit temporal modeling while preserving high-level semantic information relevant for retrieval. The choice of mean pooling reflects a trade-off between computational efficiency and temporal expressiveness.

Text Encoder and Query Representation

Textual queries are encoded using the CLIP text encoder, which maps natural language inputs into a fixed-dimensional semantic embedding space. Queries are first tokenized using CLIP's byte-pair encoding tokenizer and processed through a transformer-based text encoder to produce a global sentence-level representation.

To adapt textual representations to the characteristics of the target video dataset, the text encoder is fine-tuned while keeping the vision encoder frozen. This parameter-efficient training strategy enables improved alignment between query embeddings and precomputed video representations without incurring the high computational cost of end-to-end video–text training. Fine-tuning is performed using a symmetric contrastive loss, encouraging semantically matched query–video pairs to have higher similarity than mismatched pairs.

At inference time, each query is encoded into a normalized embedding vector, allowing efficient similarity computation against the cached video embeddings. This design supports fast retrieval while maintaining robust semantic matching across diverse query formulations.

Shared Embedding Space and Similarity Computation

The retrieval system operates in a shared multimodal embedding space learned through contrastive training. Both video and text embeddings are L2-normalized and projected into the same semantic space, enabling direct similarity-based comparison.

Given a query embedding $\mathbf{q}$ and a video embedding $\mathbf{v}$, retrieval is performed using cosine similarity. Video retrieval is performed using cosine similarity, as defined in Equation 2.3.

$$\mathrm{sim}(\mathbf{q}, \mathbf{v}) = \frac{\mathbf{q} \cdot \mathbf{v}}{\|\mathbf{q}\| \, \|\mathbf{v}\|}. \tag{2.3}$$

This similarity measure provides a computationally efficient and scalable retrieval mechanism suitable for large video collections.

During inference, similarity scores are computed between the query embedding and all cached video embeddings, and videos are ranked in descending order of similarity. This bi-encoder design avoids expensive cross-modal interaction at retrieval time, enabling low-latency querying while preserving strong semantic alignment learned during training.

2.3 Explored Design Choices

Narration-Augmented Representation

To enrich video representations with auxiliary semantic information, a narration-augmented retrieval strategy was explored during system development. This approach was motivated by recent work showing that automatically generated captions can provide complementary language-level cues that help bridge the semantic gap between visual content and natural language queries.

In this design, a short textual narration was generated offline for each video using a pretrained vision–language captioning model applied to a representative video frame. The narration serves as a coarse semantic summary of the visual content. These narrations were encoded using the same CLIP text encoder employed for query encoding, ensuring that both queries and narrations reside in a shared embedding space.

To combine visual and textual information, the visual video embedding $\mathbf{v}_{\text{video}}$ and the narration embedding $\mathbf{v}_{\text{nar}}$ were fused using a weighted linear combination. Visual and narration embeddings are combined using weighted fusion, as defined in Equation 2.4.

$$\mathbf{v}_{\text{fused}} = \alpha \mathbf{v}_{\text{video}} + (1 - \alpha) \mathbf{v}_{\text{nar}}, \tag{2.4}$$

followed by L2 normalization. The fusion weight α controls the relative contribution

of visual and narration features. Retrieval was then performed using cosine similarity between the fused video embeddings and query embeddings.

While this approach introduced richer semantic context and improved interpretability in some cases, its effectiveness was found to be highly sensitive to narration quality and relevance. Inaccurate or overly generic narrations occasionally introduced noise, leading to degraded retrieval performance. Due to this sensitivity and the additional preprocessing overhead, the narration-augmented representation was not retained in the final system configuration. This explored design is illustrated in Figure 2.1.

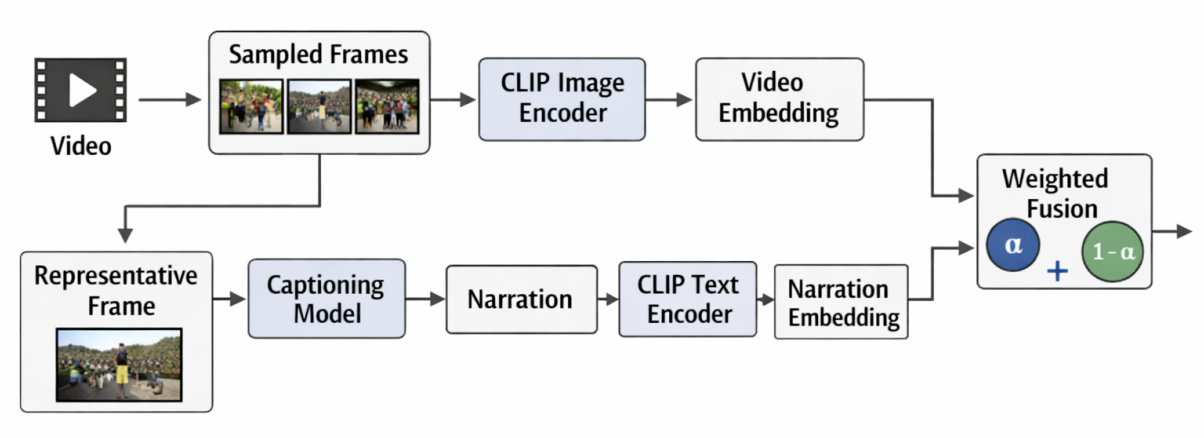

Figure 2.1: Explored narration-augmented video representation, where auxiliary textual narrations are fused with visual video embeddings in a shared CLIP embedding space.

Two-Stage Retrieval with Reranking

A two-stage retrieval strategy was also investigated to improve retrieval precision at higher ranks. This design was inspired by coarse-to-fine retrieval paradigms, which aim to balance efficiency and expressive cross-modal matching.

In the first stage, an initial set of candidate videos was retrieved using standard embedding-based similarity search in the shared CLIP embedding space. This stage leverages the efficiency of bi-encoder retrieval and is suitable for large-scale indexing. In the second stage, the top-ranked candidates were reranked using a cross-encoder model that jointly processes the query text and auxiliary video narration. Unlike bi-encoder similarity, the cross-encoder enables direct token-level interaction between query and narration, allowing for more fine-grained semantic matching.

The reranking model outputs refined relevance scores for each candidate, which are used to reorder the retrieved results. Although this approach improved semantic discrimination among top-ranked videos, it significantly increased computational cost, as cross-encoder inference must be performed for each query–candidate pair. Additionally, the effectiveness of reranking depended on the availability and quality of narrations. Given these limitations and the efficiency-oriented goals of the system, the two-stage reranking strategy was not adopted in the final architecture. The explored reranking pipeline is illustrated in Figure 2.2.

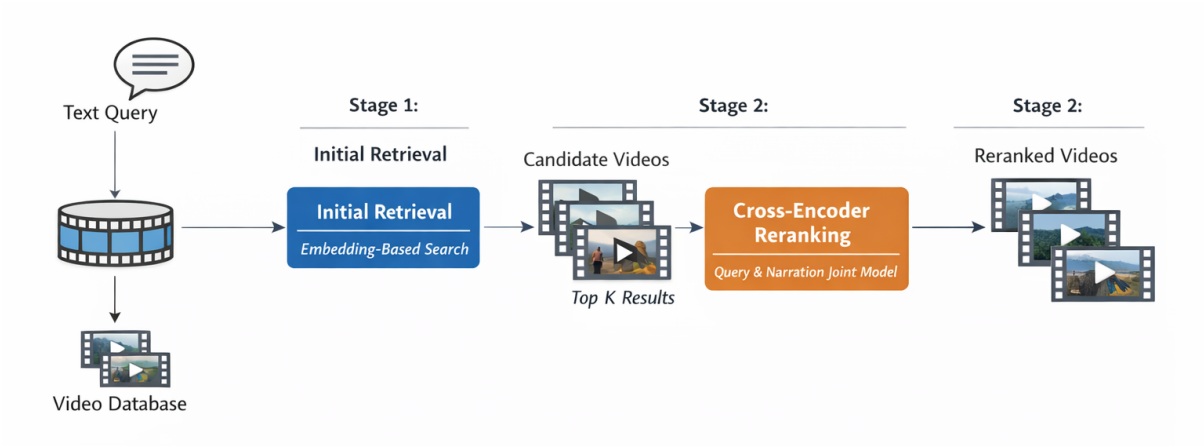

Figure 2.2: Two-stage retrieval pipeline consisting of initial embedding-based retrieval followed by cross-encoder reranking of top candidates.

Text-Only Fine-Tuning in a Dual-Encoder Framework

During system development, different training strategies were explored to improve alignment between textual queries and video representations within a dual-encoder retrieval framework. In particular, the effect of fine-tuning the text encoder while keeping the vision encoder frozen was investigated. This parameter-efficient strategy allows the model to adapt textual representations to the target dataset without incurring the high computational cost of end-to-end video–text training.

In this design, video embeddings are precomputed and cached using a frozen CLIP vision encoder, while the CLIP text encoder is fine-tuned using a symmetric contrastive loss. This training objective encourages matched text–video pairs to have high cosine similarity while pushing apart mismatched pairs in the shared embedding space. By updating only the text encoder, the system preserves efficient inference while improving query–video alignment.

Compared to more complex fusion and reranking approaches, this strategy offers a favorable trade-off between performance improvement, training stability, and computational efficiency, and therefore informs the final system configuration. The explored text-only fine-tuning strategy within the dual-encoder framework is illustrated in Figure 2.3.

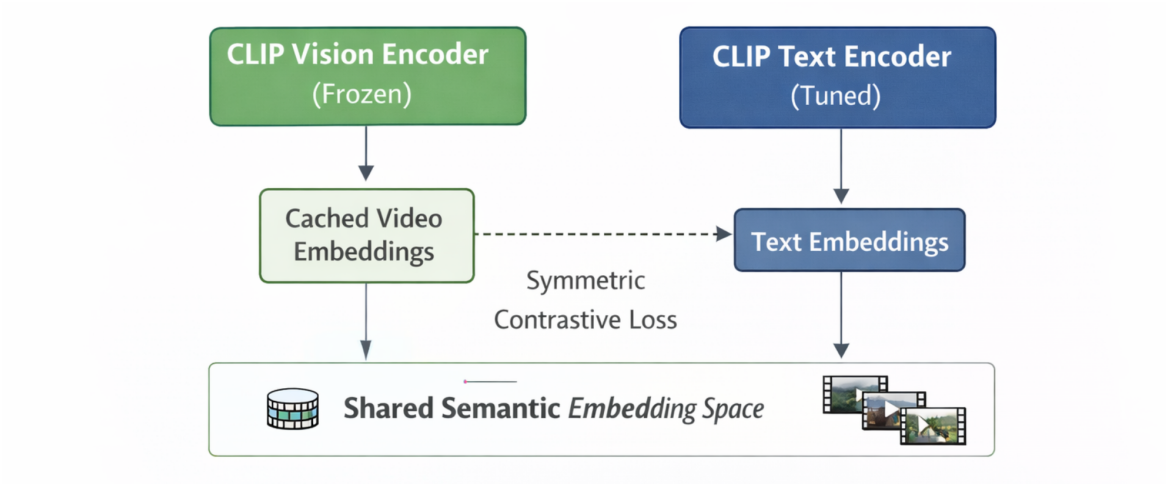

Figure 2.3: Text-only fine-tuning strategy explored during system design, where video embeddings are cached and the text encoder is fine-tuned to improve cross-modal alignment within a dual-encoder framework.

2.4 Training Strategy

The system is designed to minimize training complexity while leveraging the strong semantic alignment provided by pretrained models. The CLIP image encoder is kept frozen throughout all experiments, and video embeddings are computed once and reused for both training and evaluation. This design significantly reduces computational overhead and ensures consistent visual representations.

To improve alignment between textual queries and visual video embeddings, the CLIP text encoder is fine-tuned using a symmetric contrastive learning objective. During training, matched text–video pairs are encouraged to have higher cosine similarity than mismatched pairs within a batch. Video embeddings are treated as fixed targets, allowing training to focus exclusively on adapting textual representations to the target dataset.

No task-specific fine-tuning is performed for the reranking model. Instead, a pretrained cross-encoder is directly applied during inference to rescore top-ranked candidates. This choice avoids additional training complexity while still enabling effective semantic refinement through joint query–candidate processing.

Overall, the training strategy reflects a parameter-efficient design that prioritizes stability, reproducibility, and scalability under limited computational resources.

2.5 Implementation Details

The system is implemented in Python using the PyTorch framework. Pretrained CLIP models are loaded via the OpenCLIP library, with the Vision Transformer (ViT)-B/16 architecture used for both image and text encoders. Video frames are decoded and uniformly sampled at fixed intervals, ensuring consistent temporal coverage across videos.

Each sampled frame is independently encoded using the CLIP image encoder, and frame-level embeddings are aggregated using mean pooling to produce a single video-level representation. All video embeddings are L2-normalized and stored in a FAISS index configured for inner-product similarity, which is equivalent to cosine similarity for normalized vectors.

Text queries are tokenized using CLIP's byte-pair encoding tokenizer and encoded using the CLIP text transformer. Query embeddings are normalized before similarity computation. For reranking, the system applies a pretrained transformer-based cross-encoder that jointly processes the query text and candidate video metadata to produce refined relevance scores.

To ensure reproducibility and efficient evaluation, all embeddings are cached to disk, and evaluation is conducted on a fixed subset of queries. The implementation prioritizes modularity, allowing individual components such as retrieval and reranking to be evaluated independently.

2.6 Final System Configuration

The final system adopts a two-stage text-to-video retrieval architecture designed to balance retrieval efficiency and semantic precision. The first stage performs large-scale candidate retrieval using visual representations derived from a pretrained CLIP image encoder, while the second stage refines the ranking of top candidates using a cross-encoder reranking model.

In the initial retrieval stage, each video is represented by a fixed-length embedding obtained through uniform frame sampling and mean pooling of CLIP image features. Query text is encoded independently using the CLIP text encoder, and retrieval is performed using cosine similarity between query and video embeddings. All video embeddings are precomputed and stored in a FAISS index, enabling efficient similarity search over the full dataset.

The second stage operates on the top-K candidates retrieved in the first stage. Instead of relying solely on embedding similarity, a pretrained transformer-based cross-encoder jointly processes the query text and each candidate video representation to compute refined relevance scores. This reranking step enables deeper semantic interaction between modalities, allowing the system to correct mis-rankings caused by coarse embedding similarity.

Figure 2.4 provides an overview of the complete text-to-video retrieval pipeline implemented in this work. Given a natural language query, the query is first encoded using the CLIP text encoder to obtain a fixed-dimensional query embedding. In parallel, video embeddings are precomputed offline by uniformly sampling frames from each video, encoding them using the CLIP image encoder, and aggregating frame-level representations through mean pooling. These video embeddings are indexed using FAISS to enable efficient similarity search.

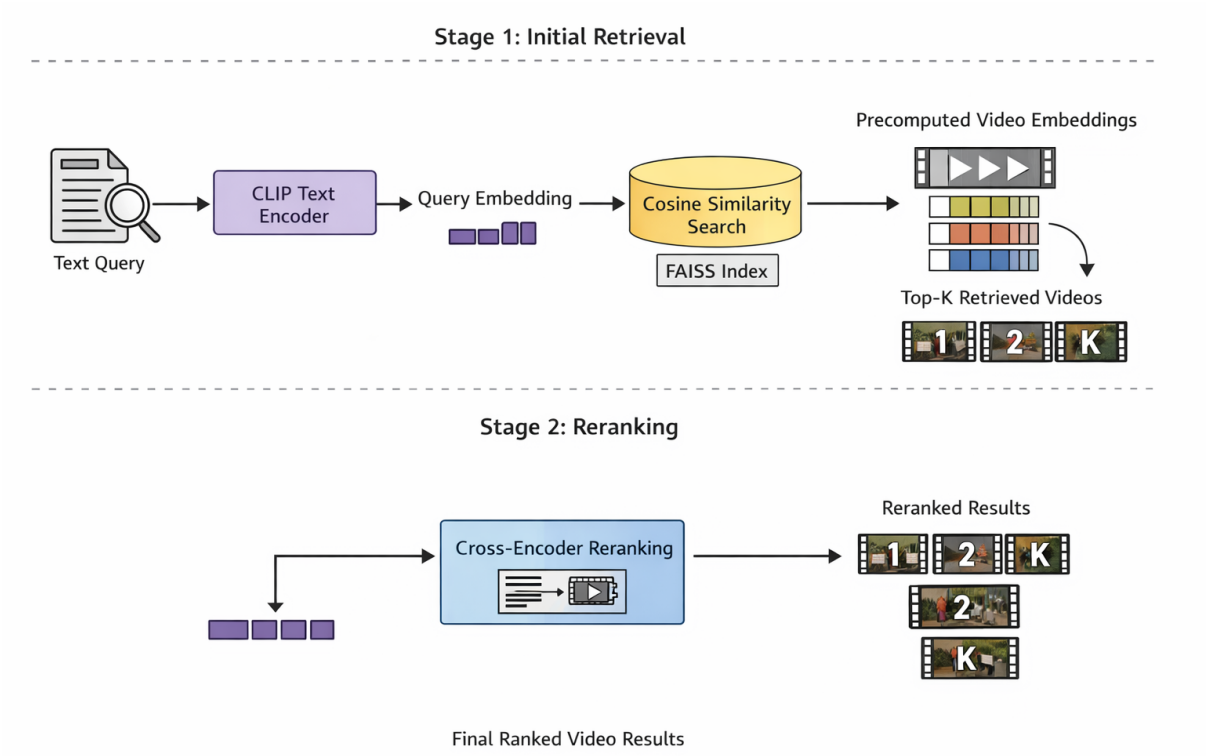

Figure 2.4: Overall system pipeline for text-to-video retrieval.

In the first retrieval stage, cosine similarity is computed between the query embedding and all indexed video embeddings to retrieve a set of top-K candidate videos. In the second stage, the retrieved candidates are refined using a cross-encoder reranking model that jointly processes the query text and each candidate video representation to produce refined relevance scores. This two-stage design combines the efficiency of embedding-based retrieval with the improved semantic discrimination offered by reranking, resulting in modest but consistent performance gains.

Auxiliary narration-based representations were explored during development but did not consistently improve retrieval performance. Consequently, the final system relies exclusively on visual embeddings for candidate generation and uses reranking as the primary mechanism for improving retrieval precision at higher ranks.

Chapter 3

Experimental Results and Discussion

This chapter presents the experimental evaluation of the proposed text-to-video retrieval system on the MSR-VTT dataset. The experiments assess the performance of a CLIP-based visual retrieval baseline, examine the effect of incorporating auxiliary design choices such as narration-based representations and dual-encoder fine-tuning, and analyze the impact of a two-stage retrieval strategy with reranking. All system configurations are evaluated under consistent experimental settings to enable fair numerical comparison across explored approaches and with representative methods reported in prior work.

3.1 Evaluation Measures

The performance of the text-to-video retrieval system is evaluated using standard metrics commonly adopted in video–language retrieval benchmarks.

Recall@K (Recall at K (R@K)) measures the proportion of text queries for which at least one relevant video appears within the top-K retrieved results. Higher Recall@K values indicate stronger retrieval accuracy. In this work, Recall is reported for $K = \{1, 5, 10\}$, capturing both early and mid-rank retrieval performance.

Median Rank (Median Rank (MedR)) reports the median rank position of the first relevant video across all text queries. Lower values correspond to better early-rank retrieval behavior.

Mean Rank (Mean Rank (MeanR)) computes the average rank position of the first relevant video across all queries. Although sensitive to outliers, Mean Rank provides insight into the overall ranking distribution.

Mean Average Precision (Mean Average Precision (mAP)) evaluates the overall quality of the ranked retrieval results by averaging precision values at the ranks where relevant videos occur. This metric captures both ranking order and retrieval precision.

Retrieval Time measures the average time required to retrieve and rank videos for a single query and is reported to assess system efficiency and scalability, particularly when comparing single-stage and two-stage retrieval strategies.

3.2 Experimental Setup

Experiments are conducted on the MSR-VTT test split. Video representations are constructed by uniformly sampling eight frames per video and encoding each frame using a pretrained CLIP image encoder. Frame-level embeddings are aggregated using mean pooling to obtain a single video-level representation.

Text queries are encoded using the CLIP text encoder. For configurations involving fine-tuning, the vision encoder remains frozen while the text encoder is optimized using a symmetric contrastive loss to improve alignment between textual queries and visual video representations. Video embeddings are precomputed and indexed using FAISS to enable efficient large-scale similarity search.

Evaluation is performed using subsets of 1,000 and 10,000 queries, depending on the experiment, to balance computational feasibility with statistical reliability.

3.3 Results of Explored and Final Configurations

Table 3.1 summarizes the retrieval performance of all system configurations explored during development. These include an initial prototype pipeline, the visual-only baseline, narration-augmented retrieval, two-stage retrieval with reranking, and dual-encoder text fine-tuning.

Table 3.1: Comparison of explored and final system configurations on MSR-VTT (1,000 queries).

Method	R@1	R@5	R@10	MedR	MeanR	mAP	Time (s)
Visual Only (Baseline)	**0.222**	0.476	**0.573**	7	100.484	**0.346**	**0.002**
Visual + Rerank (K=20)	0.206	**0.483**	0.567	**6**	100.439	0.337	0.008
Visual + Rerank (K=50)	0.207	0.481	0.571	6	N/A	0.337	0.021
Initial Prototype	0.140	0.300	0.381	24	315.4364	0.221	0.001
Visual + Narration	0.175	0.422	0.517	10	115.464	0.300	0.002
Dual Encoder (Text Fine-Tuning)	0.140	0.300	0.381	24	319.5362	0.221	0.001

While the visual-only baseline achieves the strongest overall performance in terms of Recall@1, Recall@10, mAP, and retrieval efficiency, the introduction of a reranking stage yields improvements in specific early-rank metrics. In particular, reranking consistently reduces the median rank of the first relevant video and improves Recall@5, indicating more stable ranking behavior among the top retrieved candidates. These gains, however, are accompanied by increased inference latency and do not translate into improvements at Recall@1 or overall mAP. This highlights a trade-off between ranking refinement and computational efficiency, rather than a universal performance improvement.

The initial prototype demonstrates limited retrieval accuracy and is included to document early system validation rather than serve as a competitive baseline. The visual-only baseline significantly outperforms this initial configuration across all metrics, confirming the effectiveness of CLIP-based visual representations combined with simple temporal aggregation.

Narration-augmented retrieval degrades performance relative to the visual baseline, indicating that automatically generated narrations introduce semantic noise when their content does not align with query intent. Dual-encoder text fine-tuning improves efficiency but does not yield performance gains over the visual baseline under the evaluated setting.

3.4 Effect of Two-Stage Retrieval with Reranking

To further refine retrieval performance at higher ranks, a two-stage retrieval strategy incorporating reranking was evaluated. In this configuration, the visual-only baseline is first used to retrieve a set of top-K candidate videos. These candidates are then reranked using a cross-encoder model that jointly processes the query text and candidate representations, enabling more fine-grained semantic comparison.

As shown in Table 3.1, reranking improves Median Rank from 7 to 6 and yields small gains in Recall@5. Increasing the reranking depth from $K = 20$ to $K = 50$ does not produce substantial accuracy improvements but significantly increases inference time. This behavior highlights the diminishing returns of deeper reranking and underscores the trade-off between retrieval precision and computational efficiency.

3.5 Comparison with Prior Work

Text-to-video retrieval has been extensively studied in the literature, particularly following the emergence of CLIP-based transfer methods that treat videos as sampled frames and perform retrieval in a shared embedding space. Representative approaches such as CLIP4Clip [5] demonstrate that uniform frame sampling combined with CLIP embeddings and simple aggregation can already yield strong performance, establishing a widely adopted baseline design. Subsequent methods improve upon this baseline by introducing additional components, such as temporal refinement modules (e.g., CLIP2Video [6]), multi-grained alignment objectives (e.g., X-CLIP [7]), and efficiency-oriented architectural changes (e.g., CenterCLIP [8]) or alignment/matching distillation (e.g., CLIP2TV [9]). More recent works also explore semantic enrichment through generated captions or narration-like auxiliary text signals (e.g., NarVid [12]), often combined with query-aware filtering or improved fusion mechanisms.

Compared to these prior directions, the proposed system intentionally prioritizes simplicity and compute efficiency. It follows the established CLIP transfer paradigm of frame sampling and mean pooling, with retrieval performed via cosine similarity and scalable FAISS indexing. Under this lightweight design, the visual-only baseline already achieves strong retrieval accuracy on MSR-VTT in the conducted experiments, indicating that CLIP pretraining provides robust cross-modal alignment even without explicit temporal modeling.

The explored narration-augmented representation did not improve performance in the current setup, suggesting that the usefulness of auxiliary generated text is highly dependent on narration quality and its semantic match to benchmark queries. In contrast, the two-stage reranking approach aligns with coarse-to-fine retrieval ideas: it modestly

improves early-rank behavior (e.g., Median Rank) by applying a stronger cross-encoder only to a limited candidate set, at the cost of additional inference time. Overall, the results support the conclusion that, under limited compute budgets, carefully implemented CLIP baselines can be highly competitive, while reranking offers incremental gains when higher precision is worth the latency trade-off.

3.6 Discussion

The experimental results lead to several key observations. First, the strong performance of the visual-only baseline confirms that CLIP-based representations combined with simple temporal aggregation are sufficient for competitive text-to-video retrieval, even in the absence of explicit temporal modeling.

Second, auxiliary design choices such as narration-based representations and dual-encoder text fine-tuning do not consistently improve retrieval accuracy, highlighting the sensitivity of such approaches to noise and dataset-specific characteristics.

Third, reranking provides incremental improvements in early-rank accuracy but introduces non-negligible computational overhead. These findings emphasize a clear trade-off between retrieval precision and system efficiency.

3.7 Summary

Overall, the proposed system achieves strong text-to-video retrieval performance on MSR-VTT using a lightweight CLIP-based retrieval pipeline with simple temporal aggregation and efficient FAISS indexing. The visual-only baseline provides the best accuracy–efficiency trade-off in the evaluated configurations. A two-stage reranking strategy yields modest but consistent improvements in early-rank behavior, at the expense of increased retrieval time. These results demonstrate that practical, compute-efficient CLIP transfer pipelines can perform reliably under realistic resource constraints, and that coarse-to-fine reranking is a viable refinement when additional latency is acceptable.

Chapter 4

Conclusion and Limitations

4.1 Conclusion

This work investigated the design and evaluation of a CLIP-based text-to-video retrieval system on the MSR-VTT benchmark, with an emphasis on balancing retrieval accuracy, semantic precision, and computational efficiency. Starting from a lightweight visual-only baseline, several architectural and training strategies were explored, including narration-augmented video representations, two-stage retrieval with cross-encoder reranking, and parameter-efficient text-only fine-tuning within a dual-encoder framework.

Experimental results demonstrate that simple yet carefully implemented CLIP transfer pipelines can achieve strong retrieval performance without explicit temporal modeling or heavy architectural modifications. Uniform frame sampling combined with mean pooling of CLIP image embeddings provides a robust and efficient video representation, yielding competitive Recall and mAP values under constrained computational budgets. This confirms prior findings that large-scale vision–language pretraining enables effective cross-modal alignment even when videos are treated as unordered frame collections.

Auxiliary design choices were found to offer limited or inconsistent benefits in the evaluated setting. Narration-augmented representations, while conceptually appealing for enriching semantic context, were highly sensitive to narration quality and relevance, often introducing noise that degraded retrieval performance. Similarly, text-only fine-tuning within a dual-encoder framework improved training efficiency but did not produce clear gains over the strong visual baseline.

A two-stage retrieval strategy with cross-encoder reranking yielded modest but consistent improvements in early-rank behavior, particularly in Median Rank and Recall@5. However, these gains came at the cost of increased inference latency, highlighting a clear trade-off between retrieval precision and system efficiency. Overall, the results suggest that under realistic resource constraints, prioritizing efficient embedding-based retrieval provides the most favorable accuracy–efficiency trade-off, while reranking serves as an optional refinement when additional computational cost is acceptable.

4.2 Challenges and Limitations

Despite the encouraging results, several challenges and limitations were encountered during system development and evaluation.

First, the overall task complexity of text-to-video retrieval imposes significant computational demands, particularly when operating at the scale of MSR-VTT. Processing videos requires decoding, frame sampling, and feature extraction using large pretrained vision transformers, all of which are Graphics Processing Unit (GPU)-intensive. As a result, experiments were constrained by available GPU memory and compute capacity.

Second, these hardware limitations restricted the extent of training and hyperparameter exploration. It was not feasible to substantially increase the number of training epochs, batch size, or the number of sampled frames per video without exceeding memory limits or incurring prohibitive training time. Consequently, certain design choices, such as deeper temporal sampling or more aggressive fine-tuning schedules, could not be fully explored.

Third, computational cost also limited the scalability of more expressive retrieval strategies. Two-stage retrieval with cross-encoder reranking significantly increases inference time, as cross-encoder evaluation must be performed for each query–candidate pair. While effective for refining top-ranked results, this approach does not scale efficiently to large candidate sets under constrained resources.

Finally, storage capacity posed an additional limitation. Caching visual embeddings for all videos, along with optional auxiliary representations such as narrations, requires substantial disk space. Storing both visual and textual features for multiple experimental variants further increased storage overhead, constraining the number of configurations that could be retained and evaluated concurrently.

These limitations highlight practical challenges inherent to large-scale multimodal retrieval systems. Addressing them would likely require access to more powerful computational infrastructure, more efficient model architectures, or adaptive feature compression strategies. Nevertheless, the results of this work demonstrate that even under constrained resources, carefully designed CLIP-based pipelines can deliver reliable and competitive performance on benchmark text-to-video retrieval tasks.

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Appendix A

Supplementary Materials

This appendix provides links to the source code implementations and additional resources used in this project. These materials are included for completeness and reproducibility but are not required for understanding the main body of the report.

A.1 Source Code

The full implementation of the proposed text-to-video retrieval system, including data preprocessing, model training, retrieval, and evaluation scripts, is available at the following Google Colab Notebook:

- **Google Colab Link:**
  `https://colab.research.google.com/drive/1bc1tLmsk7QtHf20S7D_lCDw-t6iFdfae?usp=sharing`

A.2 Additional Resources for Compilation

Due to the large size of video features and intermediate embeddings, some auxiliary files are hosted externally. These include:

- Precomputed video embeddings.
- Generated narration files used in explored design choices.
- Evaluation outputs and logs.

These materials are available via Google Drive:

- **Google Drive Folder:**
  `https://drive.google.com/drive/folders/1KCQEeASkqNRnecx4yY9FRn2WIz7owW71?usp=sharing`

A.3 Papers Discussed

- **Google Drive Folder:**
  https://drive.google.com/drive/folders/1Pw5yxqqPx9slyK_aK-fplXNwbNIXukaH?usp=sharing

A.4 Initial Pipeline

- **Google Colab Link:**
  https://colab.research.google.com/drive/1k1tIADoc9CXEbugvpdeq6Nry4jvOxzyY?usp=sharing

A.5 Dual-Encoding (Explored)

- **Google Colab Link:**
  https://colab.research.google.com/drive/1yCwuLbklIZxktv8FBvExFrP_r569r9WJ?usp=sharing